5 **(a)** For a progressive wave, state what is meant by:

(i) period

....

.... [1]

(ii) amplitude.

....

.... [1]

(b) A stretched string is fixed at both ends and made to vibrate with a frequency of 25 Hz. A stationary wave is formed on the string. The distance between the fixed ends of the string is 0.96 m.

Figure 5.1 shows the string at one instant.

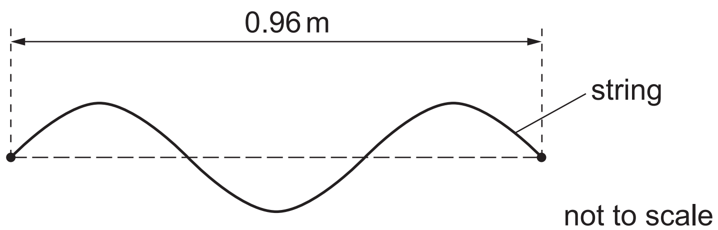

Figure 5.1

(i) On Figure 5.1, draw a cross (×) at the position of an antinode. [1]

(ii) Show that the speed of a progressive wave on the string is 16 $m s^{-1}$.

[3]

(iii) The frequency of the vibrations is increased to 30 Hz. The speed of the wave on the string does not change.

State and explain whether a stationary wave is formed.

....

....

....

.... [2]

(c) A beam of light of wavelength 4.5×10^{-7} m is incident normally on a diffraction grating with 6.7×10^{5} lines per metre. The light passes through the grating and forms a pattern of bright spots on a screen that is parallel to the grating.

Figure 5.2 shows two of the bright spots in the pattern.

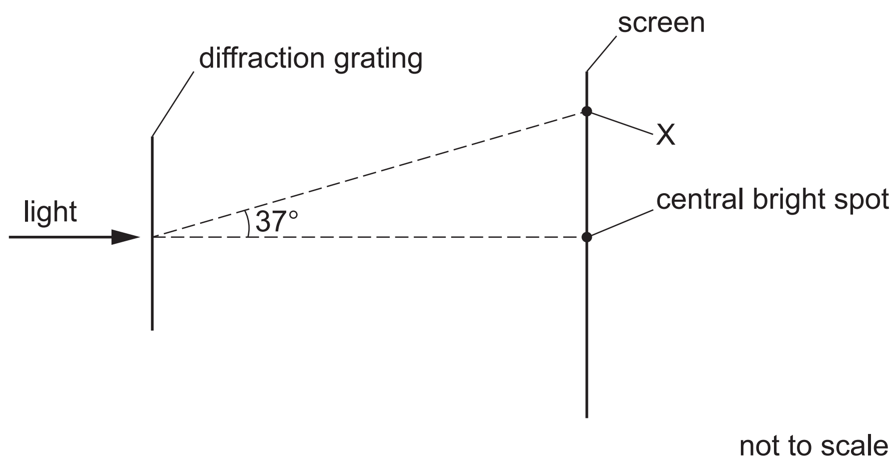

Figure 5.2

The angle formed at the diffraction grating between the central bright spot and the bright spot at position X is 37°.

(i) Calculate the order *n* of the bright spot formed at X.

n = ... [3]

(ii) The diffraction grating is replaced by a grating with fewer lines per metre.

State whether the distance between adjacent bright spots on the screen increases, decreases or remains the same.

.... [1]

[Total: 12]